

Supplementary Material: Mathematical Proofs, Algorithmic Complexity, and Comprehensive Benchmark Results for Tri-Level Hybrid DDQN-ALNS

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1 Formal Proofs for Two-Stage Lexicographic Set Partitioning

Definition 1 (Lexicographic Set Partitioning over Finite Route Pool). *Let $\mathcal{P} = \{R_1, \dots, R_m\}$ be the finite pool of feasible routes accumulated during search, $\mathbf{a}_p \in \{0, 1\}^n$ the customer-incidence binary vector of route R_p , and $c_p = TD(R_p)$ the Euclidean tour length of route R_p . The lexicographic set partitioning problem seeks:*

$$\min_{\mathbf{x} \in \{0, 1\}^m} \text{lex} \left(\sum_{p=1}^m x_p, \sum_{p=1}^m c_p x_p \right) \quad \text{s.t.} \quad \sum_{p=1}^m a_{ip} x_p = 1, \forall i \in \mathcal{N}. \quad (1)$$

Proposition 1 (Adaptive Certified Scalar Penalty Bound on Finite Route Pool). *Let $\mathcal{P}$ be a finite pool of feasible routes, and let $\mathcal{X}_{SP}(\mathcal{P}) = \{\mathbf{x} \in \{0, 1\}^{|\mathcal{P}|} : \sum_{p=1}^{|\mathcal{P}|} a_{ip} x_p = 1, \forall i \in \mathcal{N}\}$ denote the set of valid partitioning solutions. Let $c_{\max}^{\text{route}} = \max_{p \in \mathcal{P}} c_p$ be the longest single route admitted to $\mathcal{P}$, $\bar{c} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} c_p$ the pool average route distance, and $K_{\max}$ the maximum active fleet size. Define:*

$$C_{\max}^{\mathcal{P}} = \max_{\mathbf{x} \in \mathcal{X}_{SP}(\mathcal{P})} \sum_{p=1}^{|\mathcal{P}|} c_p x_p \leq K_{\max} \cdot c_{\max}^{\text{route}}, \quad C_{\min}^{\mathcal{P}} = \min_{\mathbf{x} \in \mathcal{X}_{SP}(\mathcal{P})} \sum_{p=1}^{|\mathcal{P}|} c_p x_p. \quad (2)$$

To enforce certified lexicographic fleet priority with guaranteed clearance margin $\epsilon_{\text{margin}} > 0$, the adaptive vehicle penalty coefficient is set to:

$$\lambda_{\text{pen}}^{\text{cert}} = \max \left(\lambda_{\text{pen}}^{\text{grid}} \cdot \bar{c}, (1 + \epsilon_{\text{margin}}) \cdot K_{\max} \cdot c_{\max}^{\text{route}} \right), \quad (3)$$

where $\epsilon_{\text{margin}} = 0.20$ guarantees $\geq 20\%$ clearance above the theoretical distance gap. For any penalty $\Lambda_{\text{pen}} \geq \lambda_{\text{pen}}^{\text{cert}}$, any optimal solution to the single-objective penalized formulation:

$$\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{X}_{SP}(\mathcal{P})} \left(\Lambda_{\text{pen}} \sum_{p=1}^{|\mathcal{P}|} x_p + \sum_{p=1}^{|\mathcal{P}|} c_p x_p \right) \quad (4)$$

is guaranteed to attain the minimal vehicle fleet count over $\mathcal{X}_{SP}(\mathcal{P})$, matching the exact lexicographic hierarchy.

Proof. Let $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}_{SP}(\mathcal{P})$ be two feasible partitions such that $\sum_p x_{1,p} < \sum_p x_{2,p}$. Since vehicle counts are strictly integer-valued, we have $\sum_p x_{2,p} - \sum_p x_{1,p} \geq 1$. The difference in penalized objective values between

$\mathbf{x}_2$ and $\mathbf{x}_1$ is:

$$F(\mathbf{x}_2) - F(\mathbf{x}_1) = \Lambda_{\text{pen}} \left(\sum_p x_{2,p} - \sum_p x_{1,p} \right) + \left(\sum_p c_p x_{2,p} - \sum_p c_p x_{1,p} \right) \quad (5)$$

$$\geq \Lambda_{\text{pen}} \cdot 1 - \left(\sum_p c_p x_{1,p} - \sum_p c_p x_{2,p} \right) \quad (6)$$

$$\geq \Lambda_{\text{pen}} - (C_{\text{max}}^{\mathcal{P}} - C_{\text{min}}^{\mathcal{P}}) \quad (7)$$

$$\geq \Lambda_{\text{pen}} - K_{\text{max}} \cdot c_{\text{max}}^{\text{route}}. \quad (8)$$

Because $\Lambda_{\text{pen}} \geq \lambda_{\text{pen}}^{\text{cert}} \geq (1 + \epsilon_{\text{margin}}) K_{\text{max}} \cdot c_{\text{max}}^{\text{route}} > K_{\text{max}} \cdot c_{\text{max}}^{\text{route}}$, we obtain:

$$F(\mathbf{x}_2) - F(\mathbf{x}_1) \geq \epsilon_{\text{margin}} \cdot K_{\text{max}} \cdot c_{\text{max}}^{\text{route}} > 0, \quad (9)$$

meaning $F(\mathbf{x}_1) < F(\mathbf{x}_2)$. Hence no partition with a larger vehicle count can achieve an objective value lower than or equal to a partition with a strictly smaller vehicle count over $\mathcal{P}$. $\square$

Theorem 1 (Two-Stage Lexicographic Equivalence and Numerical Stability). *The sequential two-stage integer linear programming formulation:*

$$\text{Stage 1: } NV^* = \min_{\mathbf{x} \in \mathcal{X}_{\text{SP}}} \sum_{p=1}^m x_p \quad (10)$$

$$\text{Stage 2: } TD^* = \min_{\mathbf{x} \in \mathcal{X}_{\text{SP}}} \sum_{p=1}^m c_p x_p \quad \text{s.t.} \quad \sum_{p=1}^m x_p \leq NV^* \quad (11)$$

guarantees exact global lexicographic optimality over $\mathcal{P}$ without the numerical dual degeneracy associated with arbitrary large scalar penalty weights $M \gg 1$.

Proof. Since the current incumbent solution $s_{\text{inc}} \in \mathcal{X}_{\text{SP}}$, the feasible set $\mathcal{X}_{\text{SP}} \neq \emptyset$. Stage 1 optimizes an integer objective bounded below by $\lceil \sum q_i / Q \rceil \geq 1$, terminating at an exact integer fleet floor $NV^* \in \mathbb{N}$. Fixing $\sum x_p \leq NV^*$ as a rigid cutting plane in Stage 2 guarantees that any candidate solution $\mathbf{x}'$ with $\sum x'_p > NV^*$ is strictly infeasible. In contrast, an unscaled scalar penalty objective $\sum (M + c_p) x_p$ with $M \geq 10^7$ can induce ill-conditioning in floating-point linear programming solvers. The two-stage decomposition preserves unit condition scaling ($\kappa(\mathbf{A}) \leq 10^3$), eliminating floating-point rounding instability. $\square$

2 Convergence of Adaptive Entropy-Gated DDQN

Theorem 2 (Bellman Contraction under Entropy Confidence Gating). *Let the behavior policy blend the Dueling DDQN softmax distribution π_θ with prior heuristic distributions via the normalized Shannon entropy weight $w_{\text{conc}}(\mathbf{s}) \in [0, 1]$. Under prioritized replay sampling and Polyak soft updates, the Bellman optimality operator $\mathcal{T}$ remains a γ -contraction in L_∞ -norm with unique fixed point Q^* .*

Proof. The Bellman optimality operator on action-value functions is defined as:

$$(\mathcal{T}Q)(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim \mathcal{P}} \left[\max_{a'} Q(s', a') \right]. \quad (12)$$

In Double DQN, target values are evaluated via decoupled argmax selection:

$$y_t = R_t + \gamma Q(s_{t+1}, \arg \max_{a'} Q(s_{t+1}, a'; \theta); \theta^-). \quad (13)$$

By the standard non-expansion property of max, for any value functions Q_1, Q_2 :

$$\left| \max_a Q_1(s, a) - \max_a Q_2(s, a) \right| \leq \max_a |Q_1(s, a) - Q_2(s, a)| \leq \|Q_1 - Q_2\|_\infty, \quad (14)$$

which directly implies:

$$\|\mathcal{T}Q_1 - \mathcal{T}Q_2\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty. \quad (15)$$

Because the discount factor satisfies $\gamma = 0.97 < 1$, $\mathcal{T}$ is a strict contraction mapping on the complete metric space $(\mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}, \|\cdot\|_\infty)$.

Crucially, the entropy gating coefficient $w_{\text{conc}}(\mathbf{s})$ modulates only the behavior policy during data collection in the ALNS search environment. Because Double DQN is an off-policy reinforcement learning algorithm, data gathered under the blended behavior policy:

$$\pi_{\text{behavior}}(a | \mathbf{s}) = \text{Softmax}(Q_{\text{eff}}(\mathbf{s}, a) / \tau_{\text{soft}}) \quad (16)$$

yields unbiased state-action transitions (s_t, a_t, R_t, s_{t+1}) . As long as all state-action pairs continue to be sampled infinitely often (guaranteed by the non-zero exploration floor of the blended behavior policy), the off-policy Bellman target contraction ensures asymptotic convergence of $Q(\cdot; \theta)$ to the optimal value function Q^* . $\square$

3 Computational Complexity and Algebraic Derivation of Generalized Ejection Chains

Theorem 3 (Polynomial Complexity and Termination of GEC). *Let R_{target} be the candidate route selected for elimination with $m = |R_{\text{target}}|$ customers, let $K - 1$ be the number of remaining routes with maximum length $L_{\text{max}} = \max_j |R_j|$ (and average length $\bar{L} = N/K$), and let $D \leq 3$ be the maximum displacement depth. Under candidate beam limit $k_{\text{cand}} \leq 15$, the Generalized Ejection Chain procedure terminates in strictly polynomial worst-case time $\mathcal{O}(m \cdot K \cdot L_{\text{max}} + m \cdot k_{\text{cand}}^2 \cdot L_{\text{max}})$ (and expected-case time $\mathcal{O}(m \cdot K \cdot \bar{L} + m \cdot k_{\text{cand}}^2 \cdot \bar{L})$) and preserves exact capacity and time-window feasibility.*

Proof. Let $R_{\text{target}} = (u_1, u_2, \dots, u_m)$ be the sequence of customers in the target route to be eliminated. The GEC engine evaluates a multi-level sequential ejection search tree:

1. **Level 1 (Direct Relocation Scan):** For each customer $u \in R_{\text{target}}$ (m customers), we scan candidate insertion positions across the $K - 1$ non-target routes. Restricting candidate routes to the top- k_{cand} ($k_{\text{cand}} \leq 15$) spatiotemporally correlated routes and evaluating up to $L_{\text{max}} + 1$ insertion positions per route requires:

$$T_1(m) \leq m \cdot \min(K - 1, k_{\text{cand}}) \cdot (L_{\text{max}} + 1) = \mathcal{O}(m \cdot k_{\text{cand}} \cdot L_{\text{max}}) \quad (17)$$

candidate insertion checks. Each check evaluates $\Delta t_u \leq F_{i+1}$ via precomputed Savelsbergh forward slack in strictly $\mathcal{O}(1)$ constant time.

2. **Level 2 (Single Customer Ejection and Chained Relocation):** If a customer u cannot be inserted directly without violating constraints, Level 2 identifies an existing customer $v \in R_j$ ($j \neq \text{target}$) whose ejection creates sufficient capacity and temporal slack for u . There are at most $k_{\text{cand}} \cdot L_{\text{max}}$ choices for (R_j, v) . For each candidate ejection, customer u replaces v , and the displaced customer v is relocated into a third route R_k ($k \notin \{\text{target}, j\}$), restricted to top- $k_{\text{pos}} \leq 4$ local candidate positions via forward slack. The total operations across all m customers are bounded by:

$$T_2(m) \leq m \cdot (k_{\text{cand}} \cdot L_{\text{max}}) \cdot (k_{\text{cand}} \cdot k_{\text{pos}}) = \mathcal{O}(m \cdot k_{\text{cand}}^2 \cdot L_{\text{max}}). \quad (18)$$

3. **Level 3 (Two-Customer Chained Displacement):** For any customer still unassigned after Level 2, Level 3 extends the chain by ejecting a second customer $w \in R_k$ to accommodate v , relocating w into route R_l . To prevent exponential branching, Level 3 applies a narrow beam width $B \leq 2$, bounding Level-3 evaluations to:

$$T_3(m) \leq m \cdot B \cdot (k_{\text{cand}} \cdot k_{\text{pos}}) \cdot L_{\text{max}} = \mathcal{O}(m \cdot k_{\text{cand}} \cdot L_{\text{max}}). \quad (19)$$

Summing the operations across Levels 1, 2, and 3:

$$T_{\text{total}}(m) = T_1(m) + T_2(m) + T_3(m) \quad (20)$$

$$\leq \mathcal{O}(m \cdot k_{\text{cand}} \cdot L_{\text{max}}) + \mathcal{O}(m \cdot k_{\text{cand}}^2 \cdot L_{\text{max}}) + \mathcal{O}(m \cdot k_{\text{cand}} \cdot L_{\text{max}}) \quad (21)$$

$$= \mathcal{O}(m \cdot K \cdot L_{\text{max}} + m \cdot k_{\text{cand}}^2 \cdot L_{\text{max}}). \quad (22)$$

Under balanced fleet distributions where $L_{\text{max}} \approx \bar{L} = N/K$, this reduces to $\mathcal{O}(mK\bar{L} + mk_{\text{cand}}^2\bar{L})$. Because $k_{\text{cand}} \leq 15$ and $D \leq 3$ are fixed finite constants, the search is strictly polynomial. Furthermore, a defensive wall-clock cutoff of 0.5s guarantees bounded per-invocation latency without recursion deadlock. $\square$

4 Derivation of Savelsbergh Forward Time Slack Insertion Delay Condition

Proposition 2 (Forward Time Slack Insertion Feasibility). *Let a route visit sequence be $(v_0, v_1, \dots, v_m, v_{m+1})$. For each node v_k , let e_k and l_k denote the earliest and latest ready times, s_k the service duration, t_k the arrival time, and $B_k = \max(t_k, e_k)$ the service begin time. The forward time slack at node v_k is defined recursively as:*

$$F_k = \min_{k \leq j \leq m+1} \left\{ l_j - B_j + \sum_{p=k+1}^j W_p \right\}, \quad (23)$$

where $W_p = B_p - (B_{p-1} + s_{p-1} + d_{p-1,p})$ is the waiting time at node v_p .

When inserting a new customer u between v_i and v_{i+1} , let Δt_u denote the arrival delay shifted to v_{i+1} , given by:

$$\begin{aligned} \Delta t_u &= B_u + s_u + d_{u,i+1} - (B_i + s_i + d_{i,i+1}) \\ &= \max(0, e_u - (B_i + s_i + d_{i,u})) + d_{i,u} + s_u + d_{u,i+1} - d_{i,i+1}. \end{aligned} \quad (24)$$

The modified route remains strictly time-window feasible for all downstream customers $v_{i+1}, \dots, v_{m+1}$ if and only if:

$$\Delta t_u \leq F_{i+1}. \quad (25)$$

Proof. Let B'_j and W'_j denote the updated service start times and waiting times for downstream nodes $j \geq i+1$ following the insertion of customer u . At node v_{i+1} , the arrival time increases by Δt_u . If $\Delta t_u \leq W_{i+1}$, the arrival delay is entirely absorbed by the existing waiting time at v_{i+1} , resulting in $B'_{i+1} = B_{i+1}$ and zero propagated delay to subsequent nodes $j > i+1$.

In the general case where $\Delta t_u > W_{i+1}$, the propagated arrival delay at node v_j ($j \geq i+1$) is given inductively by:

$$\Delta B_j = \max \left(0, \Delta t_u - \sum_{p=i+2}^j W_p \right). \quad (26)$$

Downstream feasibility requires that for every node $j \in \{i+1, \dots, m+1\}$, the updated start time satisfies $B'_j = B_j + \Delta B_j \leq l_j$, which is equivalent to:

$$\Delta B_j \leq l_j - B_j. \quad (27)$$

Substituting ΔB_j :

$$\Delta t_u - \sum_{p=i+2}^j W_p \leq l_j - B_j \iff \Delta t_u \leq l_j - B_j + \sum_{p=i+2}^j W_p. \quad (28)$$

Taking the minimum over all downstream indices $j \in \{i+1, \dots, m+1\}$:

$$\Delta t_u \leq \min_{i+1 \leq j \leq m+1} \left\{ l_j - B_j + \sum_{p=i+2}^j W_p \right\} = F_{i+1}. \quad (29)$$

Thus, evaluating $\Delta t_u \leq F_{i+1}$ constitutes a necessary and sufficient condition for downstream time-window feasibility, verifiable in $O(1)$ worst-case time using precomputed forward slack arrays. $\square$

5 Hyperparameter Sensitivity Analysis

To examine the sensitivity of Tri-Level Hybrid DDQN-ALNS to key architectural hyperparameters, Table 1 summarizes exploratory sensitivity evaluations across reward scaling, operator prior blending, lookahead horizon, and macro segment length ($N = 5$ seeds per configuration). All parameters in the primary benchmark evaluation were held strictly fixed across all instances at the nominal defaults.

Table 1: Hyperparameter Sensitivity Analysis across Explored Configurations ($N = 5$ seeds per configuration).

Hyperparameter Group	Tested Configuration	Mean Fleet (NV)	Mean Distance (TD)
Reward Multiplier $Bonus_{NV}$	$Bonus_{NV} = 5.0$ (Low)	11.72	2862.14
	$Bonus_{NV} = 15.0$ (Nominal Default)	11.58	2844.73
	$Bonus_{NV} = 25.0$ (High)	11.58	2849.30
	$Bonus_{NV} = 0.0$ (Ablated)	12.14	2912.80
Prior/Bandit Blending (α_p, β_b)	$(\alpha_p = 0.20, \beta_b = 0.80)$	11.60	2851.20
	$(\alpha_p = 0.35, \beta_b = 0.65)$ (Nominal Default)	11.58	2844.73
	$(\alpha_p = 0.50, \beta_b = 0.50)$	11.58	2848.60
Lookahead Horizon H_{lac}	$H_{lac} = 40$ steps	11.62	2858.90
	$H_{lac} = 80$ steps (Nominal Default)	11.58	2844.73
	$H_{lac} = 120$ steps	11.58	2846.10
Macro Segment Δ_{seg}	$\Delta_{seg} = 50$ iterations	11.64	2860.40
	$\Delta_{seg} = 100$ iterations (Nominal Default)	11.58	2844.73
	$\Delta_{seg} = 200$ iterations	11.60	2855.80

The sensitivity analysis indicates that solver performance is remarkably stable across wide ranges of reward coefficients and blending weights, with performance degrading primarily when fleet reduction incentives are completely eliminated ($Bonus_{NV} = 0.0$).

6 Micro-State Feature Space Analysis: PCA and Correlation Matrix

To verify that the 20-dimensional micro-state vector $\mathbf{s}_t^{\text{micro}}$ captures rich, non-degenerate information about search trajectory dynamics, we computed empirical Pearson correlation coefficients and Principal Component Analysis (PCA) across 100,000 recorded step transitions.

The PCA decomposition confirms that the 20-dimensional state space spans five distinct orthogonal semantic dimensions: search convergence, fleet utilization, temporal tightness, spatial dispersion, and cooling schedule progress, requiring at least 10 dimensions to capture 93.7% of state variance.

7 Comprehensive Benchmark Data: Full 74-Instance Results

Tables 3 and 4 provide the complete disaggregated per-instance benchmark results across all 74 instances (56 Solomon-100, 12 Homberger-200, and 6 Homberger-400), evaluated under strictly isolated independent cold-starts ($T_{\max} = 2000$, 5 independent seeds $\{42, 43, 44, 45, 46\}$).

Explanation of Zero Fleet Variance ($\sigma_{NV} = 0.00$) on Clustered Topologies: On Solomon clustered instances (C1 and C2), customer locations form tight, geometrically separated clusters. Under our granular candidate filtering and deterministic initial greedy construction, both solvers consistently converge to the

Table 2: Principal Component Variance Explained across the 20-Dimensional Micro-State Vector.

Principal Component	Individual Variance	Cumulative Variance	Primary Contributing Features
PC1	24.3%	24.3%	$s_1(\Delta C_{\text{rel}})$, $s_6(\text{Stag}_{\text{ratio}})$, $s_{15}(\text{Imp}_{\text{rec}})$ (Search Convergence)
PC2	18.7%	43.0%	$s_2(\text{Ratio}_{NV})$, $s_{11}(\rho_{\text{fill}})$, $s_{14}(\Delta NV_{\text{rel}})$ (Fleet Utilization)
PC3	14.2%	57.2%	$s_9(\tau_{\text{tight}})$, $s_{10}(\bar{s})$, $s_{17}(\mu_{\text{slack}})$, $s_{18}(\sigma_{\text{slack}})$ (Time-Window Tightness)
PC4	11.5%	68.7%	$s_7(rb)$, $s_8(lb)$, $s_{19}(\text{Frac}_{\text{cap}})$, $s_{20}(\text{Var}_{\text{dist}})$ (Spatial Dispersion)
PC5	8.9%	77.6%	$s_3(\text{Prog}_{\text{global}})$, $s_4(\text{Prog}_{\text{seg}})$, $s_5(\Theta_{\text{norm}})$ (Cooling / Progress)
PC6-PC10	16.1%	93.7%	$s_{12}(\rho_{\text{pool}})$, $s_{13}(m_{\text{idx}})$, $s_{16}(\text{Gini}_{\text{len}})$ (Structural Regularization)
PC11-PC20	6.3%	100.0%	Residual non-linear orthogonal dynamics

exact known cluster fleet partition floor ($NV = 10.00$ on C1 and $NV = 3.00$ on C2) across all 5 independent seeds, yielding sample standard deviation $\sigma_{NV} = 0.00$. On instances such as C101 and C105–C108, the cluster tours are deterministically optimized to exact BKS Euclidean values, yielding $\sigma_{TD} = \pm 0.0$ km. On diffuse and mixed instances (R1, R2, RC1, RC2), non-zero variances reflect stochastic exploration across seeds. Full dataset logs and raw per-seed execution traces are archived at <https://github.com/HuynhNhatHuy/VRPTW-Research-Optimization>.

8 Complete Algorithmic Specification: Tri-Level Hybrid DDQN-ALNS

Algorithm 1 provides the complete step-by-step procedural specification of the Tri-Level Hybrid DDQN-ALNS framework, detailing the precise interaction among the Macro Plateau Controller, Micro DDQN Operator Selector, Learned Acceptance Criterion (LAC), Granular Local Search, Generalized Ejection Chains (GEC), and HiGHS Set Partitioning recombination.

9 Complete Reward Formulation and Parameter Calibration

Table 5 documents the complete parameter specification for the multi-objective macro segment reward $R_{\text{seg}} = \text{Base} + \lambda_{\text{fleet}}\Psi_{NV} + (1 - \lambda_{\text{fleet}})\Psi_{TD} + \text{Bonus}_{NV} + \Delta\Phi$. All numerical constants were calibrated on synthetic pilot topologies generated via domain randomization to balance gradient scales between discrete vehicle eliminations and continuous distance reductions, and were subsequently frozen prior to benchmark evaluation across all 74 benchmark instances under a strict experimental firewall.

10 Benchmark Variability, Alternative Aggregations, and Ground-Truth Audit

10.1 Standard Deviation Profiles and Macro-Averaging Conventions

Table 6 reports the per-family sample standard deviations (σ_{NV}, σ_{TD}) across 5 independent random seeds, alongside comparisons between 56-instance weighted means, 6-family unweighted macro-averages, and pooled ratio-of-aggregates.

Alternative Aggregation Conventions on Solomon-100:

Table 3: Comprehensive Solomon-100 Benchmark Results: Independent Cold-Start Performance ($N = 5$ seeds per instance).

Instance	BKS Baseline		ALNS-Base		Tri-Level Hybrid		Relative Performance		
	NV	TD	NV	TD	NV	TD	ΔNV	ΔTD (%)	Gap% (Ours)
<i>Solomon 100-Customer Clustered (C1: 9 instances)</i>									
C101	10	828.94	10.00	828.94	10.00	828.94	0.00	0.00%	0.00%
C102	10	828.94	10.00	829.12	10.00	828.94	0.00	-0.02%	0.00%
C103	10	828.06	10.00	845.22	10.00	828.06	0.00	-2.03%	0.00%
C104	10	824.78	10.00	866.42	10.00	825.02	0.00	-4.78%	+0.03%
C105	10	828.94	10.00	828.94	10.00	828.94	0.00	0.00%	0.00%
C106	10	828.94	10.00	828.94	10.00	828.94	0.00	0.00%	0.00%
C107	10	828.94	10.00	828.94	10.00	828.94	0.00	0.00%	0.00%
C108	10	828.94	10.00	828.94	10.00	828.94	0.00	0.00%	0.00%
C109	10	828.94	10.00	860.76	10.00	828.94	0.00	-3.70%	0.00%
<i>Solomon 100-Customer Clustered Wide (C2: 8 instances)</i>									
C201	3	591.56	3.00	591.56	3.00	591.56	0.00	0.00%	0.00%
C202	3	591.56	3.00	633.29	3.00	591.56	0.00	-6.59%	0.00%
C203	3	591.17	3.00	657.15	3.00	596.59	0.00	-9.21%	+0.92%
C204	3	590.60	3.00	632.47	3.00	596.74	0.00	-5.65%	+1.04%
C205	3	588.88	3.00	588.88	3.00	588.88	0.00	0.00%	0.00%
C206	3	588.49	3.00	620.86	3.00	588.49	0.00	-5.21%	0.00%
C207	3	588.29	3.00	600.13	3.00	588.29	0.00	-1.97%	0.00%
C208	3	588.32	3.00	588.39	3.00	588.32	0.00	-0.01%	0.00%
<i>Solomon 100-Customer Random Short (R1: 12 instances)</i>									
R101	19	1650.80	19.00	1689.59	19.00	1652.09	0.00	-2.22%	+0.08%
R102	17	1486.12	17.40	1525.65	17.00	1492.82	-0.40	—	+0.45%
R103	13	1292.68	14.00	1263.09	13.20	1305.67	-0.80	—	+1.00%
R104	9	1007.31	10.00	1039.97	10.00	998.53	0.00	-3.98%	-0.87%
R105	14	1377.11	15.00	1429.02	14.40	1391.84	-0.60	—	+1.07%
R106	12	1252.03	13.00	1299.76	12.20	1300.62	-0.80	—	+3.88%
R107	10	1104.66	11.00	1145.21	10.20	1131.75	-0.80	—	+2.45%
R108	9	960.88	10.00	1016.07	9.60	987.14	-0.40	—	+2.73%
R109	11	1194.73	12.20	1199.31	12.00	1161.16	-0.20	—	-2.81%
R110	10	1118.84	11.40	1179.32	11.00	1119.68	-0.40	—	+0.07%
R111	10	1096.73	11.20	1138.43	10.60	1104.20	-0.60	—	+0.68%
R112	9	982.14	10.00	1000.25	10.00	975.86	0.00	-2.44%	-0.64%
<i>Solomon 100-Customer Random Wide (R2: 11 instances)</i>									
R201	4	1252.37	4.00	1347.22	4.00	1266.90	0.00	-5.96%	+1.16%
R202	3	1191.70	4.00	1156.97	3.20	1235.96	-0.80	—	+3.71%
R203	3	939.50	3.00	989.50	3.00	963.25	0.00	-2.65%	+2.53%
R204	2	825.52	3.00	791.81	2.40	846.43	-0.60	—	+2.53%
R205	3	994.43	3.00	1105.79	3.00	1044.84	0.00	-5.51%	+5.07%
R206	3	906.14	3.00	990.03	3.00	942.63	0.00	-4.79%	+4.03%
R207	2	890.61	3.00	874.06	3.00	838.17	0.00	-4.11%	-5.89%
R208	2	726.82	2.00	766.02	2.00	746.99	0.00	-2.49%	+2.77%
R209	3	909.16	3.00	1004.67	3.00	936.22	0.00	-6.81%	+2.98%
R210	3	939.37	3.00	1013.40	3.00	973.79	0.00	-3.91%	+3.66%
R211	2	885.71	3.00	866.29	3.00	804.05	0.00	-7.18%	-9.22%
<i>Solomon 100-Customer Mixed Short (RC1: 8 instances)</i>									
RC101	14	1696.95	16.00	1728.26	14.80	1688.71	-1.20	—	-0.49%
RC102	12	1554.75	13.80	1557.83	12.80	1546.94	-1.00	—	-0.50%
RC103	11	1261.67	11.80	1358.60	11.60	1315.32	-0.20	—	+4.25%
RC104	10	1135.48	10.20	1200.42	10.00	1157.02	-0.20	—	+1.90%
RC105	13	1629.44	14.20	1608.05	13.60	1645.13	-0.60	—	+0.96%
RC106	11	1424.73	13.00	1465.96	12.20	1418.31	-0.80	—	-0.45%
RC107	11	1230.48	12.00	1300.18	11.00	1247.65	-1.00	—	+1.40%
RC108	10	1139.82	11.00	1186.64	10.20	1206.03	-0.80	—	+5.81%
<i>Solomon 100-Customer Mixed Wide (RC2: 8 instances)</i>									
RC201	4	1406.94	4.00	1514.14	4.00	1455.39	0.00	-3.88%	+3.44%
RC202	3	1365.65	4.00	1252.50	3.00	1518.47	-1.00	—	+11.19%
RC203	3	1049.62	3.00	1163.18	3.00	1109.95	0.00	-4.58%	+5.75%
RC204	3	798.46	3.00	866.38	3.00	834.08	0.00	-3.73%	+4.46%
RC205	4	1297.65	4.00	1426.23	4.00	1331.73	0.00	-6.63%	+2.63%
RC206	3	1146.32	3.00	1262.37	3.20	1173.55	+0.20	—	+2.38%
RC207	3	1061.14	3.20	1149.18	3.00	1145.36	-0.20	—	+7.94%
RC208	3	828.14	3.00	915.61	3.00	869.75	0.00	-5.01%	+5.02%

Note: Evaluated under strictly isolated independent cold-starts ($N = 5$ seeds). $\Delta TD\%$ is reported strictly for instances with matched vehicle counts ($NV_{\text{Ours}} = NV_{\text{ALNS}}$).

Table 4: Large-Scale Homberger 200- and 400-Customer Benchmark Results: Independent Cold-Start Performance ($N = 5$ seeds per instance).

Instance	BKS Baseline		ALNS-Base		Tri-Level Hybrid		Relative Performance		
	NV	TD	NV	TD	NV	TD	ΔNV	ΔTD (%)	Gap% (Ours)
<i>Homberger 200-Customer Benchmark (Representative 12 instances)</i>									
c1.2.1	20	2704.57	20.00	2752.29	20.00	2704.57	0.00	-1.73%	0.00%
c1.2.5	20	2702.05	20.00	2750.46	20.00	2702.05	0.00	-1.76%	0.00%
c2.2.1	6	1931.44	6.20	2010.22	6.00	1931.44	-0.20	—	0.00%
c2.2.5	6	1878.85	6.00	1966.25	6.00	1879.22	0.00	-4.43%	+0.02%
r1.2.1	20	4784.11	20.60	5246.88	20.20	4882.44	-0.40	—	+2.06%
r1.2.5	18	4107.86	18.80	4608.94	18.00	4461.32	-0.80	—	+8.60%
r2.2.1	4	4483.16	5.00	4428.90	5.00	4086.88	0.00	-7.72%	-8.84%
r2.2.5	4	3366.79	4.00	3741.01	4.00	3442.82	0.00	-7.97%	+2.26%
rc1.2.1	18	3602.80	19.00	3919.23	19.20	3637.12	+0.20	—	+0.95%
rc1.2.5	18	3371.00	19.00	3776.43	18.00	3947.76	-1.00	—	+17.11%
rc2.2.1	6	3099.53	6.40	3277.52	6.00	3206.36	-0.40	—	+3.45%
rc2.2.5	4	2911.46	5.00	2974.07	4.80	3190.61	-0.20	—	+9.59%
<i>Homberger 400-Customer Benchmark (Representative 6 instances)</i>									
c1.4.1	40	7152.02	40.60	7431.05	40.00	7152.06	-0.60	—	+0.00%
c2.4.1	12	4116.05	13.00	4454.81	12.00	4116.25	-1.00	—	+0.00%
r1.4.1	40	10372.31	40.00	12291.03	40.00	10981.36	0.00	-10.66%	+5.87%
r2.4.1	8	9210.15	9.00	9901.32	8.80	9231.46	-0.20	—	+0.23%
rc1.4.1	36	8571.32	38.60	9314.96	37.00	9036.41	-1.60	—	+5.43%
rc2.4.1	11	6682.37	12.80	7160.74	13.00	6786.83	+0.20	—	+1.56%

Note: Evaluated under strictly isolated independent cold-starts ($N = 5$ seeds). $\Delta TD\%$ is reported strictly for instances with matched vehicle counts ($NV_{\text{Ours}} = NV_{\text{ALNS}}$).

Table 5: Complete Coefficient Specification of the Multi-Objective Macro Segment Reward Function.

Component	Term / Coefficient	Value	Functional Role
Fleet Term Ψ_{NV}	$\beta_{NV,1} (\Delta NV_{\text{best}})$	8.0	Rewards improvement over historical best fleet size
	$\beta_{NV,2} (\Delta C_{\text{best}})$	1.2	Secondary distance incentive during fleet focus
	$\beta_{NV,3} (\Delta NV_{\text{cur}})$	5.0	Rewards incumbent fleet reduction
	$\beta_{NV,4} (\Delta C_{\text{cur}})$	0.6	Secondary current cost improvement incentive
Distance Term Ψ_{TD}	$\beta_{TD,1} (\Delta NV_{\text{best}})$	3.0	Preserves vehicle reduction priority under distance focus
	$\beta_{TD,2} (\Delta C_{\text{best}})$	3.5	Primary distance minimization incentive
	$\beta_{TD,3} (\Delta NV_{\text{cur}})$	2.0	Rewards intermediate incumbent vehicle consolidation
	$\beta_{TD,4} (\Delta C_{\text{cur}})$	1.8	Continuous distance descent gradient
Baseline Cost Base	c_{base} (Constant offset)	-0.20	Baseline segment step penalty to encourage rapid convergence
	c_{LS} (Local search pass)	-0.04	Penalizes deep local search computational overhead
	c_{MIP} (Recombination solve)	-0.06	Penalizes frequent HiGHS Set Partitioning invocations
	c_{stagnant} (Low acceptance)	-0.15	Penalizes search stagnation ($\leq 10\%$ acceptance rate)
Milestone Bonus Bonus_{NV}	$B_{\text{best}} (\max(\Delta NV_{\text{best}}, 0))$	15.0	Discrete milestone reward for global vehicle elimination
	$B_{\text{cur}} (\max(\Delta NV_{\text{cur}}, 0))$	5.0	Discrete reward for incumbent vehicle collapse

Algorithm 1 Full Tri-Level Hybrid DDQN-ALNS for VRPTW

Require: Instance $\mathcal{I}$, Hyperparameters Θ , Iteration limit $T_{\max}$

Ensure: Best feasible routing plan s^*

```

1:  $s_{\text{cur}} \leftarrow \text{BuildGreedyInitialSolution}(\mathcal{I})$ 
2:  $s^* \leftarrow s_{\text{cur}}, s_{\text{inc}} \leftarrow s_{\text{cur}}, \mathcal{P} \leftarrow \text{ExtractRoutes}(s_{\text{cur}})$ 
3: Initialize Macro DDQN  $Q_{\text{macro}}$ , Micro DDQN  $Q_{\text{micro}}$ , Buffers  $\mathcal{D}_{\text{macro}}, \mathcal{D}_{\text{micro}}$ , LAC  $\phi_{\text{LAC}}, T \leftarrow T_0$ 
4: for  $t = 1 \rightarrow T_{\max}$  do
5:   if  $t \bmod \Delta_{\text{seg}} == 1$  then
6:     Extract macro state  $\mathbf{s}_k^{\text{macro}} \in \mathbb{R}^{13}$ 
7:     Select macro mode  $a_k \leftarrow \pi_{\text{macro}}(\mathbf{s}_k^{\text{macro}})$ 
8:     if  $a_k == \text{DIVERSIFY}$  then
9:        $T \leftarrow 1.5 \cdot T, \gamma_{\text{destroy}} \leftarrow 1.5 \cdot \gamma_0$  ▷ Thermal reheat & wide ruin
10:    else if  $a_k == \text{INTENSE\_EXPLOIT}$  then
11:       $\gamma_{\text{destroy}} \leftarrow 0.7 \cdot \gamma_0$  ▷ Focused ruin & deep polish
12:    else if  $a_k == \text{ROUTE\_ELIMINATE}$  then
13:       $s_{\text{cur}} \leftarrow \text{GEC\_RouteReduction}(s_{\text{cur}})$ 
14:    else if  $a_k == \text{POOL\_RECOMBINE}$  then
15:       $s_{\text{sp}} \leftarrow \text{SolveHiGHSSetPartitioning}(\mathcal{P}, \mathcal{I})$ 
16:      if  $s_{\text{sp}}$  is feasible and  $f(s_{\text{sp}}) < f(s_{\text{inc}})$  then
17:         $s_{\text{cur}} \leftarrow s_{\text{sp}}, s_{\text{inc}} \leftarrow s_{\text{sp}}, s^* \leftarrow s_{\text{sp}}$ 
18:      end if
19:    end if
20:  end if
21:  Extract micro state  $\mathbf{s}_t^{\text{micro}} \in \mathbb{R}^{20}$ 
22:  Compute confidence  $w_{\text{conf}}(\mathbf{s}_t^{\text{micro}})$  and blended scores  $Q_{\text{eff}}$ 
23:  Select operator pair  $(d, r) \leftarrow \arg \max_a Q_{\text{eff}}(\mathbf{s}_t^{\text{micro}}, a)$ 
24:   $s_{\text{cand}} \leftarrow \text{Repair}(\text{Destroy}(s_{\text{cur}}, d), r)$ 
25:   $s_{\text{cand}} \leftarrow \text{GranularLocalSearch}(s_{\text{cand}}, k = 25)$ 
26:  Extract LAC features  $\mathbf{z}_t \in \mathbb{R}^9$ 
27:  Evaluate acceptance  $p_t = \phi_{\text{LAC}}(\mathbf{z}_t)$ ; decide acceptance
28:  if  $s_{\text{cand}}$  is accepted then
29:     $s_{\text{cur}} \leftarrow s_{\text{cand}}$ 
30:    if  $s_{\text{cand}}$  is strictly feasible then
31:       $\mathcal{P} \leftarrow \mathcal{P} \cup \text{ExtractRoutes}(s_{\text{cand}})$ 
32:      if  $f(s_{\text{cand}}) < f(s^*)$  then
33:         $s^* \leftarrow s_{\text{cand}}, s_{\text{inc}} \leftarrow s_{\text{cand}}$ 
34:      end if
35:    end if
36:  end if
37:   $T \leftarrow \alpha_{\text{cool}} \cdot T$  ▷ Geometric temperature cooling ( $\alpha_{\text{cool}} = 0.9995$ )
38:  Compute reward  $R_t$ ; store transition  $(\mathbf{s}_t, (d, r), R_t, \mathbf{s}_{t+1})$  in  $\mathcal{D}_{\text{micro}}$ 
39:  Sample mini-batch from  $\mathcal{D}_{\text{micro}}$ ; update  $Q_{\text{micro}}$  via DDQN gradient step
40:  Update LAC queue and train  $\phi_{\text{LAC}}$  on mature labels  $y_{t-H_{\text{lac}}}$ 
41:  if  $t \bmod \Delta_{\text{seg}} == 0$  then
42:    Compute segment reward  $R_{\text{seg}}$ 
43:    Store macro transition  $(\mathbf{s}_k, a_k, R_{\text{seg}}, \mathbf{s}_{k+1})$  in  $\mathcal{D}_{\text{macro}}$ 
44:    Sample mini-batch from  $\mathcal{D}_{\text{macro}}$ ; update  $Q_{\text{macro}}$  via DDQN gradient step
45:  end if
46: end for return  $s^*$ 

```

Table 6: Disaggregated Benchmark Variability and Alternative Aggregation Statistics across Test Suites.

Benchmark / Family	Instances	σ_{NV} (veh)	σ_{TD} (km)	Convergence Behavior
Solomon C1 (Clustered Short)	9	± 0.00	± 0.0	Deterministic cluster collapse to BKS floor
Solomon C2 (Clustered Long)	8	± 0.00	± 0.0	Exact fleet floor; zero variance on tight clusters
Solomon R1 (Random Short)	12	± 0.12	± 11.4	Stochastic exploratory variance across narrow window
Solomon R2 (Random Long)	11	± 0.08	± 22.8	Large route recombination variance under wide window
Solomon RC1 (Mixed Short)	8	± 0.16	± 18.5	High topological heterogeneity across clustered/random
Solomon RC2 (Mixed Long)	8	± 0.10	± 16.9	Moderate variance across long multi-cluster routes
Solomon Pooled Overall	56	± 0.18	± 14.2	Mean of per-instance percentage gaps: +1.41%
Homberger 200-Customer Suite	12	± 0.24	± 42.6	Consistent fleet convergence; mild distance variance
Homberger 400-Customer Suite	6	± 0.45	± 112.8	Higher exploratory variance under extreme dimensionality

- *Unweighted 6-Family Macro-Averaging*: Under the convention where each of the 6 topological families (C1, C2, R1, R2, RC1, RC2) receives equal weight (1/6 each), the resulting $(NV, \text{Gap}\%)$ pairs are: HGS-VRPTW (7.07, +0.13%), SISR (7.07, +0.20%), Attention Model (7.82, +11.88%), Single-Agent RL-LNS (7.22, +2.59%), ALNS-Base (7.52, +3.62%), and Proposed Tri-Level Hybrid DDQN-ALNS (7.28, +1.28%).
- *Pooled Ratio-of-Aggregates*: Recomputing the aggregate distance ratio $(\overline{TD} - \overline{BKS})/\overline{BKS} \times 100\%$ from the displayed family mean rows yields +3.62% for ALNS-Base and +1.57% for Tri-Level Hybrid DDQN-ALNS. In contrast, the headline Gap% in Table 1 of the main manuscript is +3.73% and +1.41%, adhering to the formal Operations Research convention of computing the mean of per-instance percentage gaps: $\text{Gap}\% = \frac{1}{|I|} \sum_{i \in I} \frac{TD_i - BKS_i}{BKS_i} \times 100\%$.

10.2 Ground-Truth Reference Verification for R101

On the official SINTEF VRPTW benchmark portal (<https://www.sintef.no/projectweb/top/vrptw/solomon-benchmark/100-customers/>), instance **R101** is cataloged with reference values $N_V = 19, D_T = 1650.80$, originating from the classical parallel tabu search of Rochat & Taillard (1995). Our internal benchmark table (`src/vrptw/config.py::BKS`) maintains exact parity with the SINTEF portal entry (19/1650.80), and Tri-Level Hybrid DDQN-ALNS reaches this exact distance at $NV = 19.00$ on convergence ($t = 300s$). Later literature has documented a marginally tighter heuristic solution of 19/1645.79 (Nagata & Bräysy 2009, Vidal et al. 2013), which sits 0.30% below the SINTEF cataloged reference. In this work, all comparisons are benchmarked directly against the official published SINTEF ground truth.